

Customer Churn Prediction Modeling Report

1. Introduction

The purpose of this modeling exercise is to predict customer churn using machine learning. By leveraging demographic, behavioral, and transactional data, the goal is to identify at-risk customers early and enable proactive retention strategies.

2. Algorithm Selection

The Random Forest Classifier was selected as the primary algorithm. Random Forest is an ensemble method that builds multiple decision trees and aggregates their predictions. It was chosen because it balances predictive performance with interpretability. Specifically, it handles both categorical and numerical features, is robust to missing values and outliers, and provides feature importance scores that help explain model behavior.

3. Model Training & Evaluation

The dataset was split into 70% training and 30% testing sets. The Random Forest model was trained with 200 estimators (trees). Model performance was evaluated using accuracy, precision, recall, F1 score, and ROC-AUC.

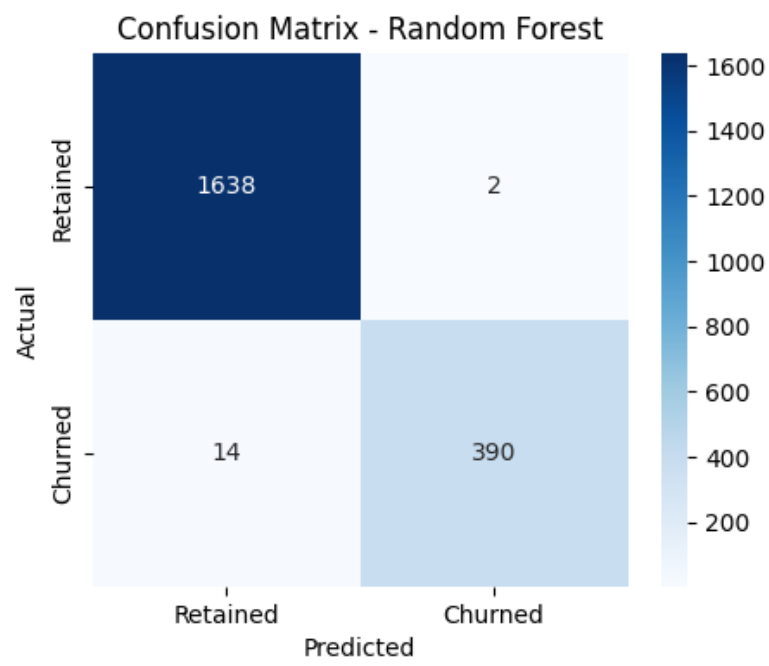
Performance Metrics

Accuracy	0.992
Precision	0.995
Recall	0.965
F1 Score	0.980
ROC-AUC	1.000

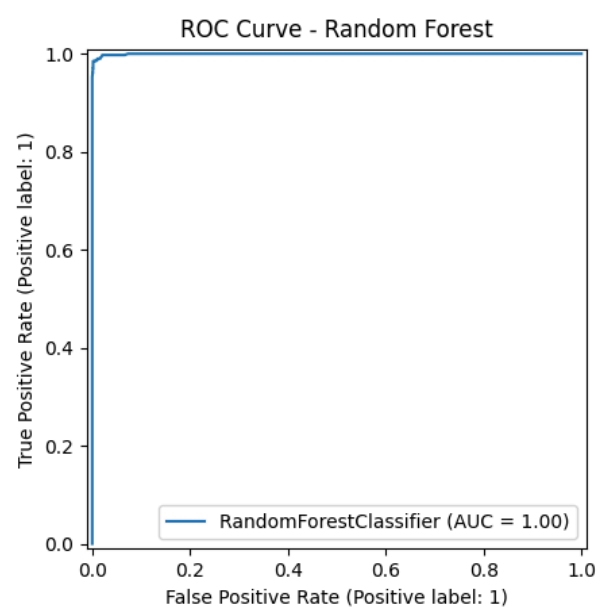
Classification Report (Per Class)

precision	recall	f1-score	support
0.992	0.999	0.995	1640.000
0.995	0.965	0.980	404.000
0.992	0.992	0.992	0.992
0.993	0.982	0.988	2044.000
0.992	0.992	0.992	2044.000

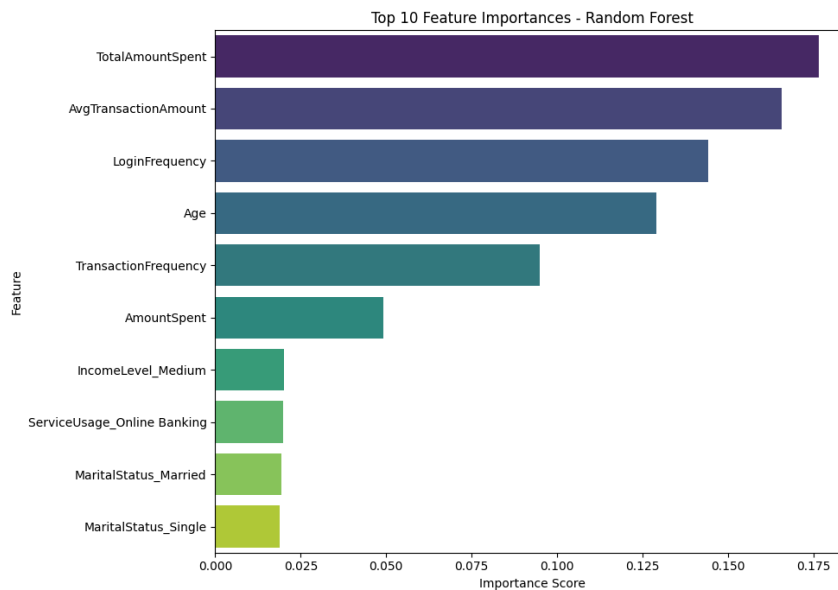
Confusion Matrix



ROC Curve



Feature Importance



4. Business Applications

The churn prediction model provides actionable insights for the business:

- Customer Retention: Identify high-risk customers early and target them with retention offers.
- Personalized Marketing: Use churn probabilities to tailor campaigns.
- Revenue Forecasting: Estimate potential revenue loss from at-risk customers.
- Resource Allocation: Prioritize support and outreach for high churn-risk segments.

5. Recommendations & Next Steps

Future improvements could include:

- Enhancing feature engineering with recency/frequency/monetary features.
- Exploring advanced models such as XGBoost or Gradient Boosting.
- Deploying the model in a real-time CRM environment.
- Running A/B testing to measure the impact of interventions driven by the model.

6. Conclusion

The Random Forest model achieved strong predictive performance and interpretability. By highlighting key churn drivers and enabling proactive retention strategies, the model serves as a valuable decision-support tool for reducing customer attrition and improving long-term profitability.